Can We Trust the Texas Marriage Index?

A Data Quality Audit

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138,500 marriages in 2020 seems low for a state of 29 million

Caitlin's instinct: Texas had roughly 50,000 abortions per year before Dobbs. If there are only 138,500 marriages, that ratio feels off.

138,584
marriages in 2020 index

29.2M
Texas population

4.7
rate per 1,000?

So we audited the index files against two independent, certified sources.

We have 13 years of individual marriage records from DSHS

Texas DSHS Marriage License Application Index, 2009–2021

- Individual-level records: names, ages, date, county
- 13 files spanning 13 years
- 4 different file formats across those 13 years
- ~2.3 million total records

Years	Format	Delimiter	Columns
2009–2011	TXT	Space/fixed-width	Husband/Wife
2012–2016	TXT	Space or asterisk	Husband/Wife
2017–2019	CSV	Comma	Husband/Wife
2020–2021	TXT	Fixed-width	Applicant 1/2

Source: Texas DSHS Vital Statistics, Marriage License Application Indexes

Two independent benchmarks let us verify every year

Benchmark 1: CDC/NCHS marriage rates $\times$ Census population

- CDC publishes state marriage rates per 1,000 population (provisional)
- Multiply by Census Bureau PEP population estimates
- Implied count $=$ rate $\times$ population / 1,000

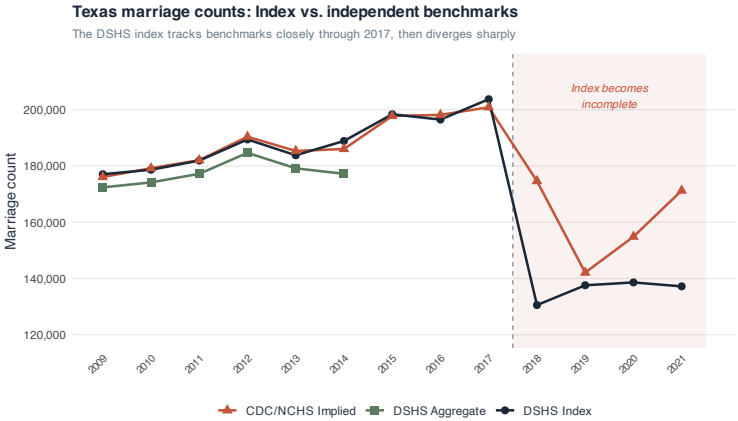
Benchmark 2: Texas DSHS aggregate totals (Table 7)

- DSHS publishes annual total marriage counts in Vital Statistics Annual Reports
- Available 2009–2014 from published reports
- These are *final* counts, not index applications

Neither benchmark uses the index files. Both are independently certified.

Source: CDC/NCHS state-marriage-rates-90-95-00-23.xlsx; Census Bureau PEP API; DSHS Annual Reports

2009–2017: The index is remarkably accurate



Sources: DSHS Marriage Index files; CDC/NCHS marriage rates × Census PEP population; DSHS Vital Statistics Annual Reports (Table 7)

Nine years of near-perfect alignment with federal benchmarks

Index vs. CDC/NCHS implied count, 2009–2017

Year	Index Count	CDC Implied Count	Δ CDC (%)	DSHS Agg. Count	Δ DSHS (%)
2009	177,084	176,093	0.6	172,395	2.7
2010	178,663	179,218	−0.3	174,171	2.6
2011	181,932	182,084	−0.1	177,219	2.7
2012	189,451	190,417	−0.5	184,690	2.6
2013	183,823	185,362	−0.8	179,173	2.6
2014	188,908	186,054	1.5	177,230	6.6
2015	198,404	197,784	0.3	—	—
2016	196,514	198,192	−0.8	—	—
2017	203,766	200,896	1.4	—	—

✔ CDC discrepancy within ±2% for every year.

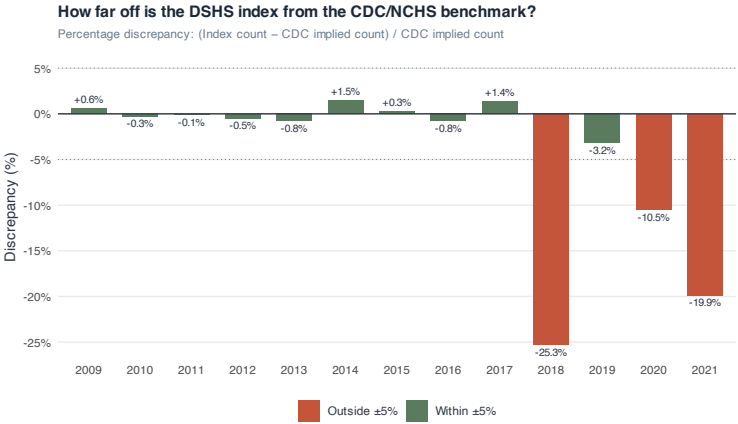
✔ Index ≈ 2.6% above DSHS aggregates—consistent with voided licenses.

2018: The index drops 36% while real marriages fall only 14%

	2017	2018	Change
Index records	203,766	130,529	−36%
CDC implied count	200,896	174,635	−13%
Discrepancy	+1.4%	−25.3%	

The 2018 file is also the first year in CSV format (`marrage2018_index.csv`). The file has 7 columns instead of the previous 8–10—the index likely became **a partial extract** after the data system changed.

The incompleteness persists through 2021



CDC implied count = CDC/NCHS state marriage rate × Census Bureau population estimate.
Negative values mean the index is missing marriages.

Four format changes in 13 years suggest an evolving data system

Period	Format	Labels	Counties	Reliable?
2009–2011	Space-delimited TXT	Husband/Wife	1 field	✓
2012–2016	Space or $*$-delimited TXT	Husband/Wife	1 field	✓
2017–2019	Comma-separated CSV	Husband/Wife	1 field	✗
2020–2021	Fixed-width TXT	Applicant 1/2	2 fields	✗

The 2020–2021 files added a second county field (license vs. marriage county) and adopted gender-neutral labels. But the key issue: the public index became **an incomplete extract** starting 2018.

A second benchmark confirms: the index counts license applications

Index vs. DSHS aggregate totals, 2009–2014

The index consistently runs ∼2.6% *above* DSHS aggregate marriage counts.

Year	Index	DSHS Agg.	Difference	Δ %
2009	177,084	172,395	+4,689	+2.7%
2010	178,663	174,171	+4,492	+2.6%
2011	181,932	177,219	+4,713	+2.7%
2012	189,451	184,690	+4,761	+2.6%
2013	183,823	179,173	+4,650	+2.6%

This ∼2.6% excess is consistent with license *applications* that do not result in recorded marriages (voided, expired, or cancelled licenses).

For our analysis: 2009–2017 is the usable window

Good news:

- The study period for HB2 effects (2009–2015) falls entirely within the reliable window
- The index matches CDC benchmarks within 2% for every year we need
- County-level counts will be accurate for the diff-in-diff design

Caution:

- 2018–2021 data are incomplete and should not be used without correction
- If we want to extend the analysis post-Dobbs, we need DSHS aggregate counts or a corrected index
- The divorce index (2009–2017) coincidentally ends exactly at the reliability boundary

Recommended next steps

1. **Proceed with 2009–2017 marriage data** for the HB2 analysis
2. **Aggregate to county-month counts** for the diff-in-diff design
3. **Contact DSHS** (`VSTAT@dshs.texas.gov`) to clarify post-2017 index incompleteness
4. **Document the audit** in the project notes for future reference and replication

The R pipeline (`code/R/01_audit_marriage_counts.R`) is fully replicable and produces all figures and tables shown in this deck.

Trust, but verify.

The data told us exactly where to draw the line.